

PREETINDERJEET SINGH

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SUMMARY

Machine Learning undergraduate experienced in building, training, and deploying deep learning models — from time-series prediction to computer vision — across defense-tech, research, and applied engineering work.

EDUCATION

Guru Nanak Dev Engineering College (GNDEC), Ludhiana

Aug 2023 – Jun 2027

B.Tech in Computer Science: 8 CGPA (Till 6th Sem)

SKILLS

Languages: Python, JavaScript

ML / DL: PyTorch, TensorFlow, Scikit-learn, Random Forest, Neural Networks, LSTM, Transformers, Transfer Learning (EfficientNet), Computer Vision, NLP, SHAP, ARIMA, Feature Engineering, Data Preprocessing, Model Training

Deployment & Tools: Docker, FastAPI, TensorFlow Lite, Edge Deployment, Model Deployment, Production Deployment, Cloud Deployment (Render, Railway), Git

Data & Retrieval: SQL, SQLite, Snowflake, Pinecone, Embeddings, RAG Pipelines, LangChain

EXPERIENCE

Machine Learning R&D Engineer — Lupex Space

Jun 2026 – Present

- Design and train LSTM, PINN, and Transformer models for missile classification and hypersonic trajectory prediction
- Lead a team of ML R&D interns, reporting directly to the co-founder, on simulation-based ML pipelines for aerospace/defense applications
- Generate, clean, and structure time-series datasets, then validate simulation outputs across runs to ensure reliable model performance

Data Science Intern / Research Fellow — Punjab AI Excellence (PAI) — Remote

Jun 2025 – Jul 2025

- Designed and trained a baseline CNN for waste classification, then fine-tuned an EfficientNet-B0 transfer-learning model under Dr. Sandeep Singh Sandha, reaching 87.08% test accuracy
- Curated and cleaned a 3,000+ image dataset, correcting class imbalance across five categories to improve generalization
- Optimized and deployed the model with TensorFlow Lite for real-time inference in smart-bin edge systems

Contractual Developer — 3S Scholars

Aug 2026 – Present

- Designing the SQL schema and building a deployment-grade attendance system end-to-end for a K-12 school, working directly with the CEO
- Implementing role-based access control, CSV export, and automated backups for administrative reporting

LEADERSHIP & INVOLVEMENT

Student Coordinator, Panel Discussions — ICCT-SD 2026 (GNDEC Ludhiana × NIT Delhi)

- Coordinated panel discussion logistics and speaker/participant interactions, and co-prepared end-to-end reports and questionnaires with faculty

PROJECTS

AI Offer Letter Agent — RAG-based document generation system

FastAPI · React · LangChain · Pinecone · Docker · Render

- Built a retrieval-augmented generation pipeline (PDF → chunk → embed → retrieve) grounding offer-letter generation in real company policy documents
- Added a second, independent LLM chain — a summarization agent — producing candidate-facing summaries of generated letters
- Resolved a CORS misconfiguration and an insecure debug-mode default, then containerized and deployed via Docker with a CPU-only PyTorch build and cached embeddings to cut cold-start latency

[Live demo — ai-offerletter-gen.onrender.com/]

Explainable AI-Based Spatio-Temporal Flood Risk Intelligence — Research (Preprint)

Random Forest · SHAP · ARIMA · Panel Regression

- Trained a Random Forest model on 640 observations, achieving 73.4% accuracy and 0.859 ROC-AUC; used SHAP analysis to identify a nonlinear deforestation threshold (~47 ha)
- Designed the study's integrated methodology, combining panel regression, Random Forest classification, SHAP explainability, and ARIMA forecasting

PUBLICATIONS

Adaptive Multi-UAV Coordination for Efficient Disaster Response Under Uncertainty — ICAAV 2026, Paper 090

Co-author — DRDO-organized conference, presented Aug 2026

- Co-authored and presented a paper applying coordination models to multi-UAV disaster-response systems under uncertainty
- Extended abstract accepted for the conference's Book of Abstracts; full paper under further review for Springer proceedings

CERTIFICATIONS

Time Series Analysis (HackerEarth × IIT Guwahati, 2026) · Python Programmer Certification (Infosys Springboard)